Skills Summary

Zeros, Multiplicity, and Graph Behavior

Use this reference while completing your worksheet or when studying.

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1. Factor first, then read the zeros

Factored form is the graph's form. A zero is a value that makes one factor zero, so the zeros are invisible until the polynomial is factored.

Order of attack: GCF $\to$ difference of squares $\to$ trinomial $\to$ quadratic in x^2.

Forgetting the GCF loses the zero at $x = 0$ — the most common lost root.

Example: Factor $x^3 - 9x$ and list its zeros.
Step 1. Every term has an x, so pull the GCF before anything else — that keeps the zero at the origin.
Step 2. $x^3 - 9x = x(x^2 - 9)$, and $x^2 - 9$ is a difference of squares.
Step 3. $\boldsymbol{x(x-3)(x+3)}$, so the zeros are $x = 0$, $x = 3$, $x = -3$.

Try it: Factor $x^3 - 2x^2 - 8x$ completely.

check: $\boldsymbol{x(x-4)(x+2)}$

2. Multiplicity: crossing, touching, flattening

Multiplicity is the exponent on a factor. It decides the shape of the curve at that zero:

- odd, equal to one — passes straight through (crosses)
- even — touches the axis and turns back, no sign change
- odd, greater than one — crosses, but flattens as it passes

The multiplicities add up to the degree.

Example: Describe the behavior of $y = (x-1)^2(x+4)$ at each zero.
Step 1. The polynomial is already factored, so read each zero and its exponent straight off rather than expanding.
Step 2. $(x-1)^2$ gives $x = 1$ with multiplicity 2 (even, so it touches); $(x+4)$ gives $x = -4$ with multiplicity 1 (odd, so it crosses).
Step 3. Zeros: $\boldsymbol{x = 1}$ (touch) and $\boldsymbol{x = -4}$ (cross); degree $2 + 1 = 3$.

Try it: Give the zeros of $y = (x+3)^2(x-2)$ and say which one touches.

check: $\boldsymbol{x = -3}$ (touch), $\boldsymbol{x = 2}$ (cross)

3. End behavior from degree and leading coefficient

Only the leading term matters once $|x|$ is large.

Even degree: both ends agree — both up if the leading coefficient is positive, both down if it is negative.

Odd degree: the ends disagree — up on the right and down on the left if the leading coefficient is positive, reversed if it is negative.

Example: Find the end behavior of $y = -x^4 + 7x$ as $x \to \infty$.
Step 1. End behavior is set by the leading term alone, so compare degree and sign instead of evaluating the whole polynomial.
Step 2. Degree 4 is even, so both ends agree; the leading coefficient is negative, so both ends fall.
Step 3. As $x \to \infty$, $y \to \boldsymbol{-\infty}$ (and the left end falls too).

Try it: For $y = 5x^3 - 2x$, what happens as $x \to -\infty$?

check: $-\infty$

4. Building a polynomial from its zeros

Run the reading backwards. A zero at $x = r$ contributes the factor $(x - r)$; how the graph behaves there sets the exponent.

Crosses $\to$ exponent 1. Touches $\to$ exponent 2.

"Least degree" means use the smallest exponent that gives the stated behavior, then multiply out.

Example: Least-degree polynomial, leading coefficient 1, that touches at $x = -1$ and crosses at $x = 4$.
Step 1. Behavior fixes each exponent: touching needs an even power, crossing needs an odd one, so take the smallest of each.
Step 2. $(x+1)^2(x-4) = (x^2+2x+1)(x-4)$, and multiplying out gives $x^3 - 2x^2 - 7x - 4$.
Step 3. $\boldsymbol{x^3 - 2x^2 - 7x - 4}$

Try it: Least degree, leading coefficient 1, touching at $x = -2$ and crossing at $x = 1$.

check: $\boldsymbol{x^3 + 3x^2 - 4}$